

Project Design Phase-II

Technology Stack (Architecture & Stack)

Project Name: ComplaintEase – Advanced Complaint Management Platform

Technical Architecture

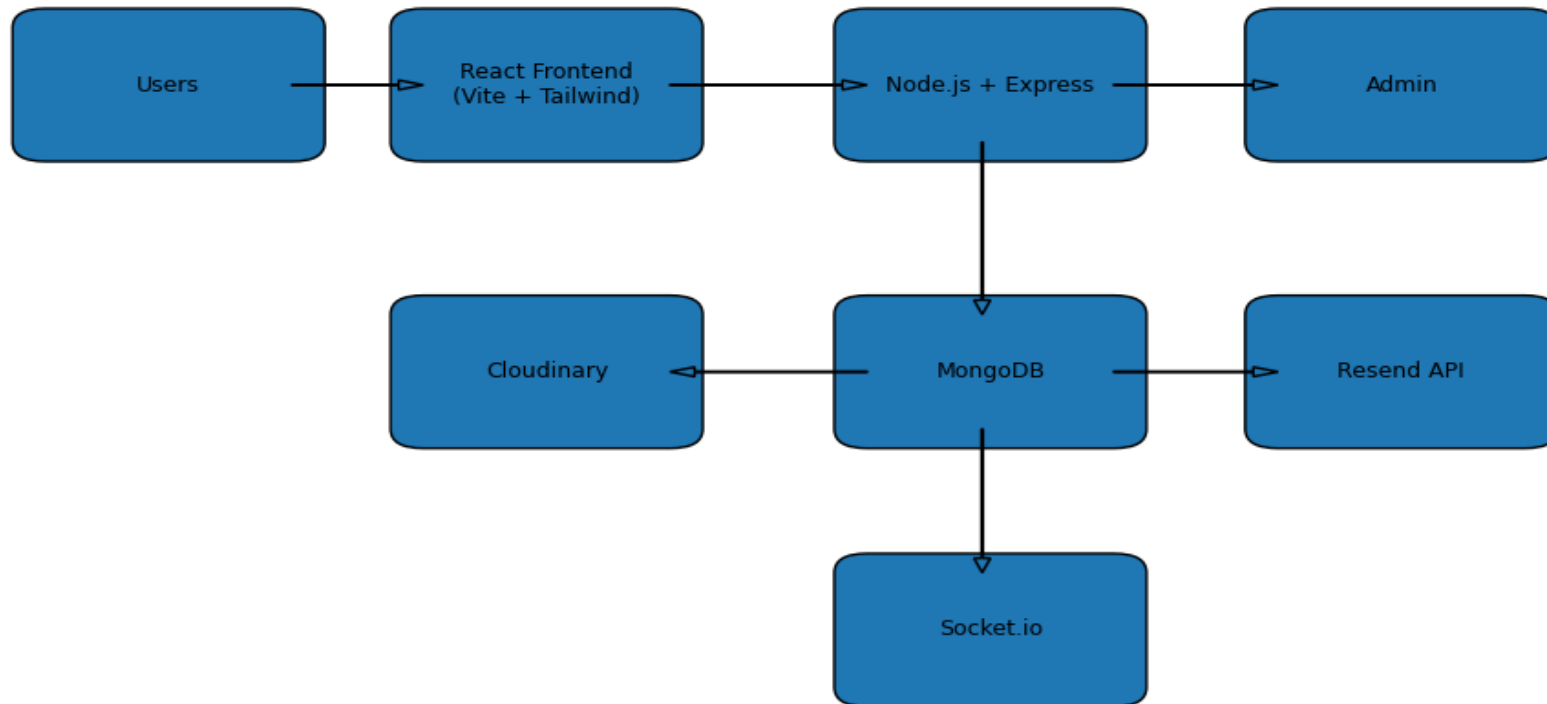


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based complaint management UI	React 18, Vite, Tailwind CSS
2	Application Logic-1	Authentication & Authorization	Node.js, Express.js, JWT
3	Application Logic-2	Complaint Management Workflow	Express Controllers, Mongoose
4	Application Logic-3	Real-time Notifications	Socket.io
5	Database	Complaint, User and Activity Data	MongoDB Atlas
6	Cloud Database	Managed Cloud Database	MongoDB Atlas
7	File Storage	Complaint Attachments	Cloudinary
8	External API-1	Email Notifications	Resend API
9	External API-2	Image/File Upload Services	Cloudinary API
10	Machine Learning Model	AI Complaint Categorization	Keyword/NLP Analyzer
11	Infrastructure	Cloud Deployment	Vercel + Render

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in application	React, Express, Node.js, Tailwind CSS
2	Security Implementations	Authentication, authorization, protection	JWT, Helmet, CORS, Rate Limiting
3	Scalable Architecture	Three-tier scalable architecture	MERN Stack, REST APIs
4	Availability	Cloud deployment with high uptime	Vercel, Render, MongoDB Atlas
5	Performance	Optimized APIs and frontend rendering	TanStack Query, Vite, MongoDB Indexing